(i) The entries in the columns corresponding to $p \vee \mathbb{T}$ and $\mathbb{T}$ are identical and hence they are equivalent. The entries in the columns corresponding to $p \vee \mathbb{F}$ and p are identical and hence they are equivalent.

Dually

(ii) $p \wedge \mathbb{T} \equiv p$ and $p \wedge \mathbb{F} \equiv \mathbb{F}$ can be proved.

6. Complement Laws

(i) $p \vee \neg p \equiv \mathbb{T}$ and $p \wedge \neg p \equiv \mathbb{F}$ (ii) $\neg \mathbb{T} \equiv \mathbb{F}$ and $\neg \mathbb{F} \equiv \mathbb{T}$

Proof

p	$\neg p$	$\mathbb{T}$	$\neg \mathbb{T}$	$\mathbb{F}$	$\neg \mathbb{F}$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F	F	T	T	F
F	T	T	F	F	T	T	F

Table 12.19

(i) The entries in the columns corresponding to $p \vee \neg p$ and $\mathbb{T}$ are identical and hence they are equivalent. The entries in the columns corresponding to $p \wedge \neg p$ and $\mathbb{F}$ are identical and hence they are equivalent.

(ii) The entries in the columns corresponding to $\neg \mathbb{T}$ and $\mathbb{F}$ are identical and hence they are equivalent. The entries in the columns corresponding to $\neg \mathbb{F}$ and $\mathbb{T}$ are identical and hence they are equivalent.

7. Involution Law or Double Negation Law

$\neg(\neg p) \equiv p$

Proof

p	$\neg p$	$\neg(\neg p)$
T	F	T
F	T	F

Table 12.20

The entries in the columns corresponding to $\neg(\neg p)$ and p are identical and hence they are equivalent.

8. de Morgan's Laws

(i) $\neg(p \wedge q) \equiv \neg p \vee \neg q$ (ii) $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Proof of (i)

p	q	$\neg p$	$\neg q$	$p \wedge q$	$\neg(p \wedge q)$	$\neg p \vee \neg q$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

Table 12.21

The entries in the columns corresponding to $\neg(p \wedge q)$ and $\neg p \vee \neg q$ are identical and hence they are equivalent. Therefore $\neg(p \wedge q) \equiv \neg p \vee \neg q$. Dually (ii) $\neg(p \vee q) \equiv \neg p \wedge \neg q$ can be proved.

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